PlantPulse

Helping operators investigate unusual plant readings

Equipment data quality shaped my approach

Available history

244

daily equipment records per plant from 1 Jan 2026 to 1 Sep 2026

62

plant performance dates per plant from Jul 2026 to Aug 2026

Findings to verify

21 in A / 10 in B

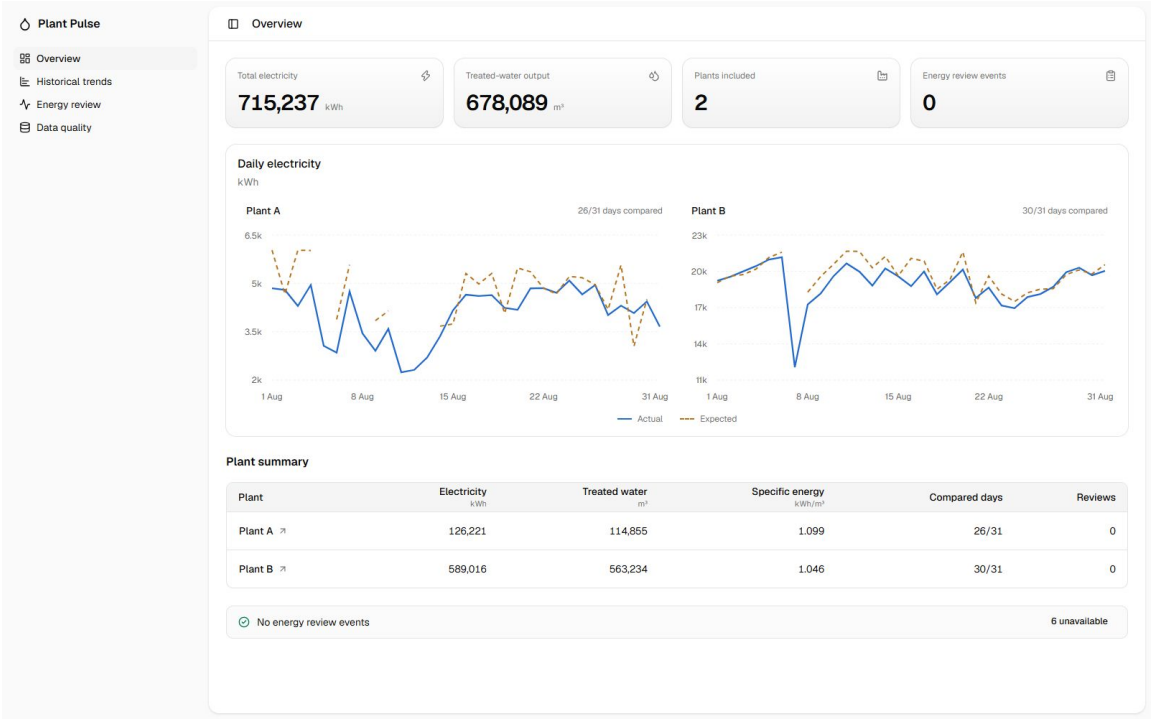
Adjacent repeated equipment records

183 day flat periods

Both Plant A runtime counters

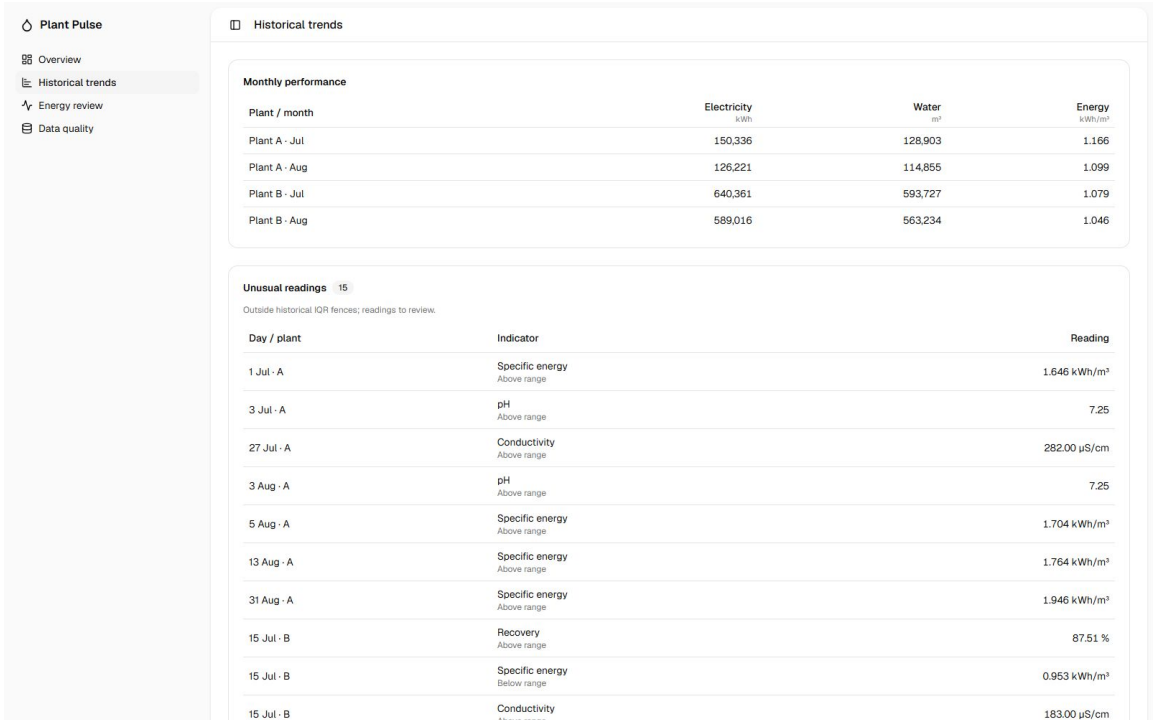
Dashboard

1. Overview
2. Historical context
3. Plant A readings
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5. Comparison coverage
6. Energy review



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High energy per cubic metre needs production context

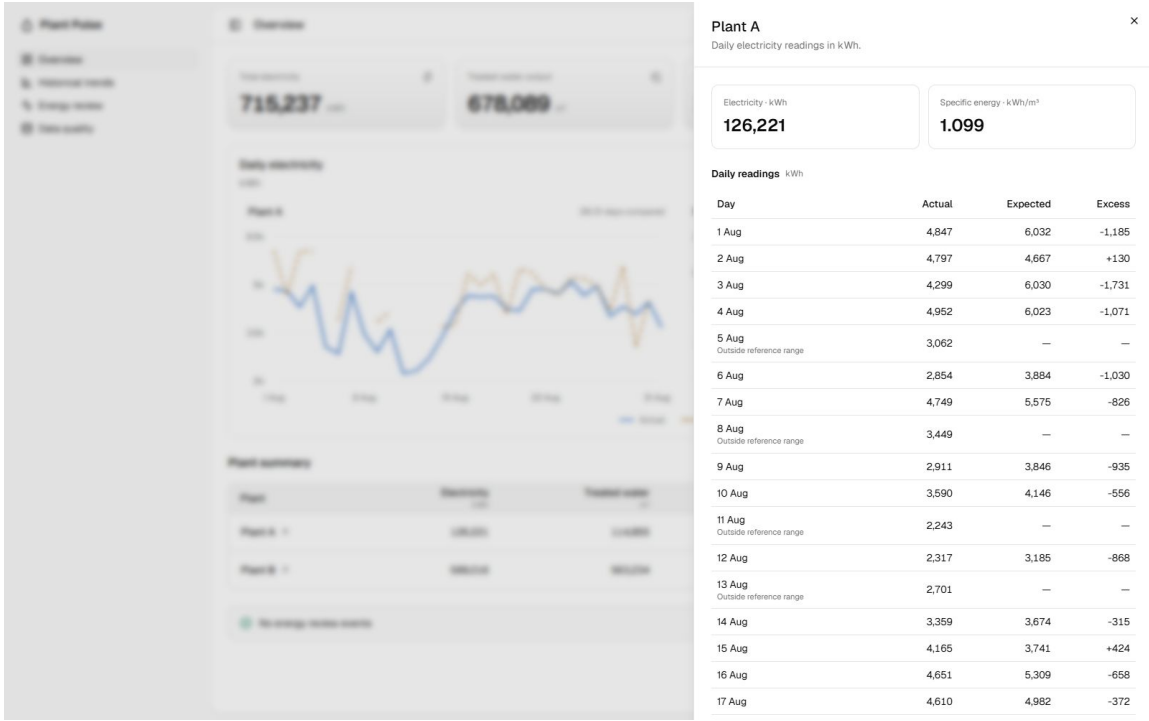
Plant A, 31 Aug 2026

3660	1881	1.946
kWh electricity	m³ treated water	kWh/m³ specific energy

Low production contributed to the high energy-per-volume ratio. Check operating conditions before assigning a cause.

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Plant A

Daily electricity readings in kWh.

Electricity · kWh	Specific energy · kWh/m³
126,221	1.099

Daily readings kWh

Day	Actual	Expected	Excess
1 Aug	4,847	6,032	-1,185
2 Aug	4,797	4,667	+130
3 Aug	4,299	6,030	-1,731
4 Aug	4,952	6,023	-1,071
5 Aug Outside reference range	3,062	—	—
6 Aug	2,854	3,884	-1,030
7 Aug	4,749	5,575	-826
8 Aug Outside reference range	3,449	—	—
9 Aug	2,911	3,846	-935
10 Aug	3,590	4,146	-556
11 Aug Outside reference range	2,243	—	—
12 Aug	2,317	3,185	-868
13 Aug Outside reference range	2,701	—	—
14 Aug	3,359	3,674	-315
15 Aug	4,165	3,741	+424
16 Aug	4,651	5,309	-658
17 Aug	4,610	4,982	-372

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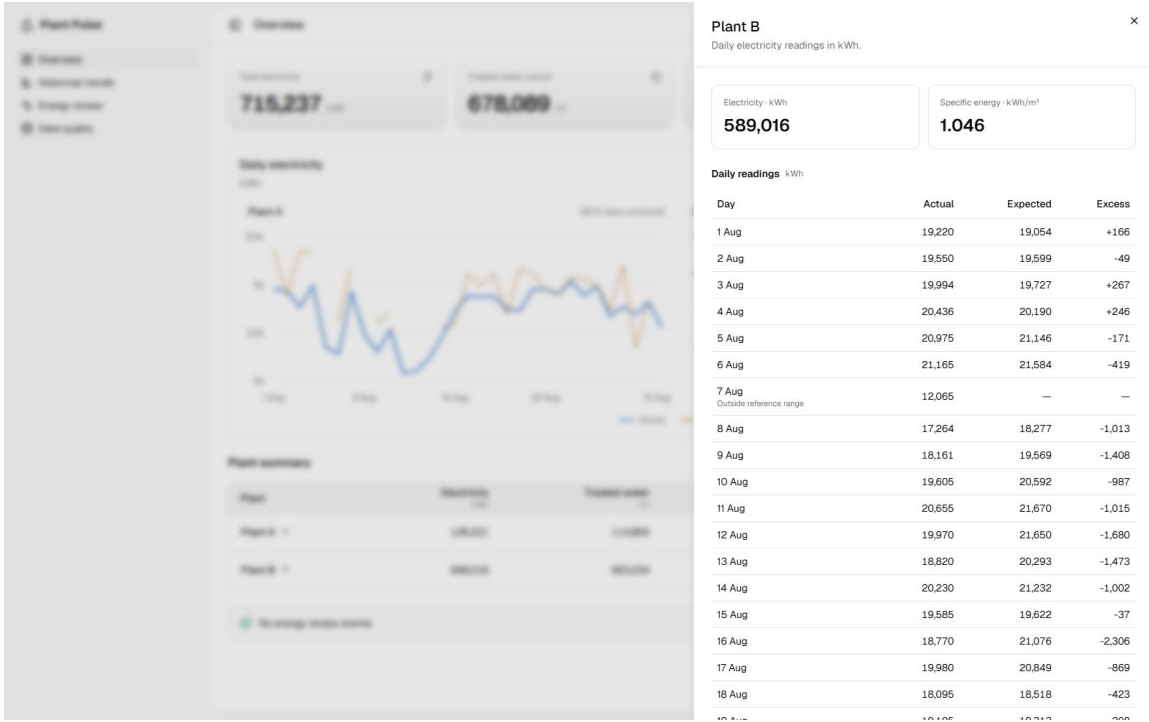
Plant A

Daily electricity readings in kWh.

9 Aug	2,911	3,846	-935
10 Aug	3,590	4,146	-556
11 Aug Outside reference range	2,243	–	–
12 Aug	2,317	3,185	-868
13 Aug Outside reference range	2,701	–	–
14 Aug	3,359	3,674	-315
15 Aug	4,165	3,741	+424
16 Aug	4,651	5,309	-658
17 Aug	4,610	4,982	-372
18 Aug	4,636	5,311	-675
19 Aug	4,239	4,070	+169
20 Aug	4,181	5,478	-1,297
21 Aug	4,846	5,361	-515
22 Aug	4,853	4,852	+1
23 Aug	4,708	4,676	+32
24 Aug	5,088	5,216	-128
25 Aug	4,659	5,177	-518
26 Aug	4,949	4,960	-11
27 Aug	4,018	4,189	-171
28 Aug	4,309	5,568	-1,259
29 Aug	4,080	3,052	+1,028
30 Aug	4,439	4,506	-67
31 Aug Outside reference range	3,660	–	–

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Plant B

Daily electricity readings in kWh.

Electricity · kWh	Specific energy · kWh/m³
589,016	1.046

Daily readings kWh

Day	Actual	Expected	Excess
1 Aug	19,220	19,054	+166
2 Aug	19,550	19,599	-49
3 Aug	19,994	19,727	+267
4 Aug	20,436	20,190	+246
5 Aug	20,975	21,146	-171
6 Aug	21,165	21,584	-419
7 Aug Outside reference range	12,065	—	—
8 Aug	17,264	18,277	-1,013
9 Aug	18,161	19,569	-1,408
10 Aug	19,605	20,592	-987
11 Aug	20,655	21,670	-1,015
12 Aug	19,970	21,650	-1,680
13 Aug	18,820	20,293	-1,473
14 Aug	20,230	21,232	-1,002
15 Aug	19,585	19,622	-37
16 Aug	18,770	21,076	-2,306
17 Aug	19,980	20,849	-869
18 Aug	18,095	18,518	-423

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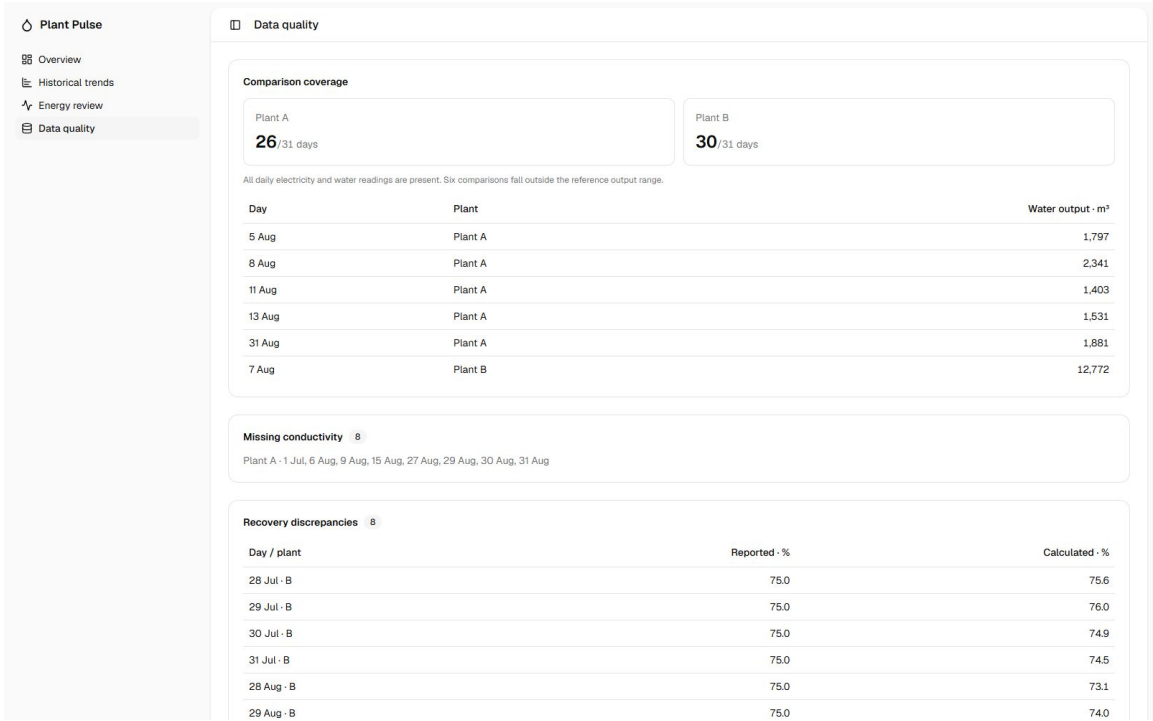
Plant B

Daily electricity readings in kWh.

8 Aug	17,264	18,277	-1,013
9 Aug	18,161	19,569	-1,408
10 Aug	19,605	20,592	-987
11 Aug	20,655	21,670	-1,015
12 Aug	19,970	21,650	-1,680
13 Aug	18,820	20,293	-1,473
14 Aug	20,230	21,232	-1,002
15 Aug	19,585	19,622	-37
16 Aug	18,770	21,076	-2,306
17 Aug	19,980	20,849	-869
18 Aug	18,095	18,518	-423
19 Aug	19,105	19,313	-208
20 Aug	20,145	21,609	-1,464
21 Aug	17,795	17,392	+403
22 Aug	18,650	19,610	-960
23 Aug	17,165	18,122	-957
24 Aug	16,950	17,504	-554
25 Aug	17,875	18,223	-348
26 Aug	18,125	18,527	-402
27 Aug	18,720	18,569	+151
28 Aug	19,925	19,729	+196
29 Aug	20,300	20,118	+182
30 Aug	19,680	19,783	-103
31 Aug	20,041	20,553	-512

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How the electricity monitor works

Jul Reference

Total electricity divided by total treated water output, separately for each plant.

Daily Comparison

Expected electricity equals the reference multiplied by today’s water output.

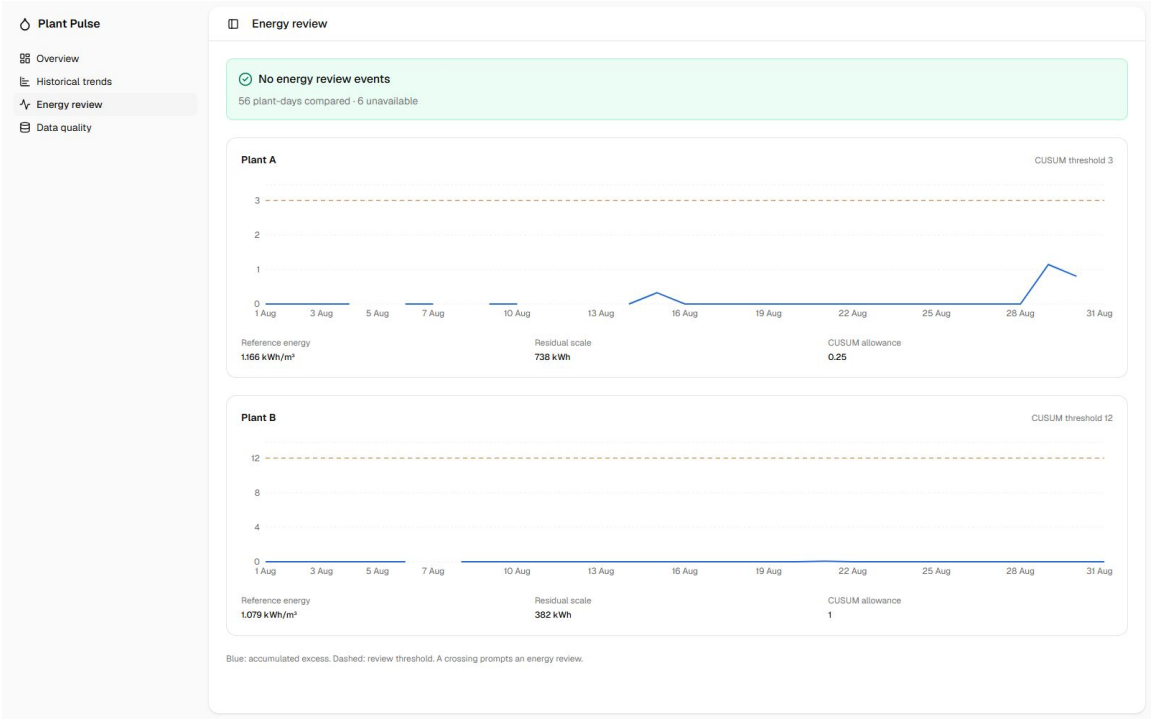
Accumulated excess

CUSUM builds positive evidence towards a provisional investigation threshold.

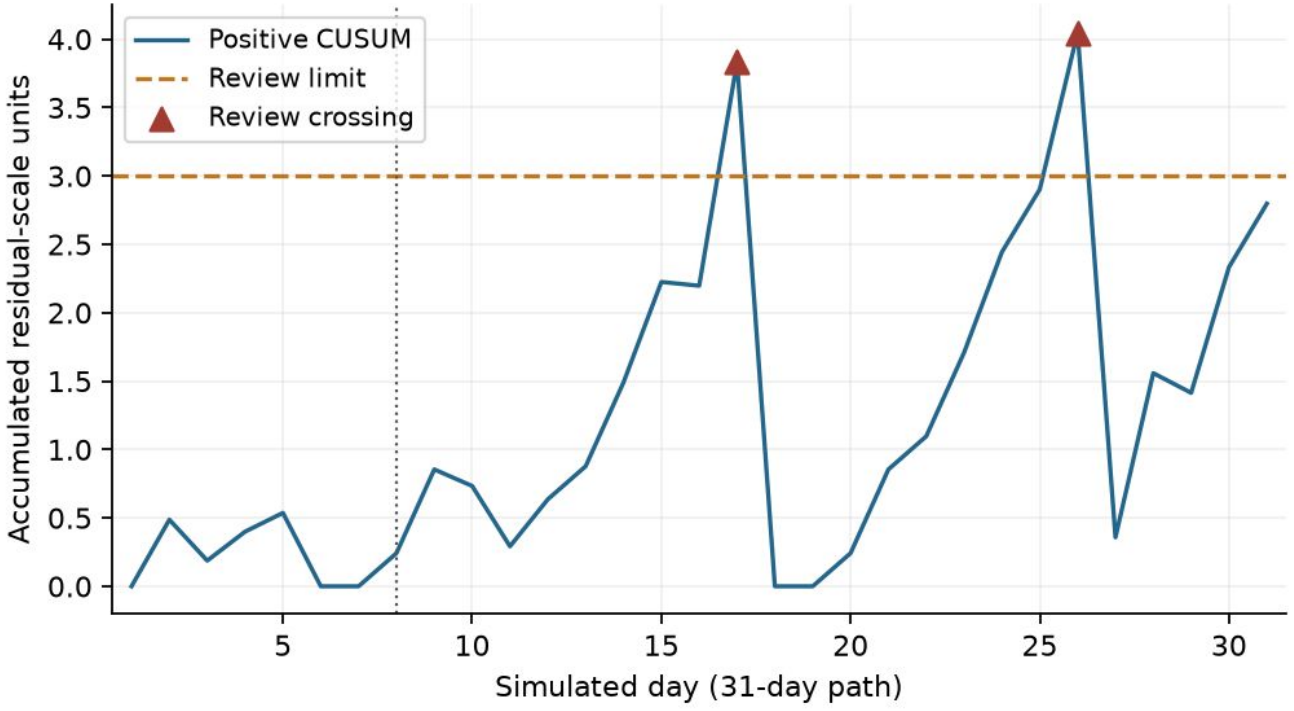
A crossing prompts operator review. An isolated spike can also trigger a crossing.

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A constructed increase triggers a review



+485 kWh/day

Added from simulated day 8

Red triangles mark review crossings. Accumulation restarts after each crossing.

Simulation detection and review workload

	Plant A	Plant B
Detected within 14 days	95.25%	100%
Median delay among detected paths	6 days	3 days
No change 31-day paths with at least one review	6.75%	7.5%

400 paths per plant per scenario. 3 day residual blocks resampled from July.

Conditional simulations. Field fault accuracy, failure lead time and savings remain unverified.

A pilot with BlueNexus operators

1

Confirm equipment signal meanings, units, aggregation and pump mappings.

2

Review dated operating, maintenance and cleaning events with operators.

3

Evaluate fresh data with agreed reference periods, review thresholds and reset procedures.

PlantPulse brings operational readings, historical context and electricity comparisons into one investigation workflow. Thank you.